ARX: A Statistical Gate for Autonomous Research Loops

(What happened when a local LLM wrote 19 experiments)

Abstract

Autonomous research loops that combine an LLM with tool access are usually evaluated on how much work they complete. We ask a different question: of the claims such a loop produces, how many survive a check on whether the measurement could have resolved them at all?

We built ARX, a loop whose reports are gated by classical measurement theory rather than by the loop's own judgement. The gate refuses a claim when the gold producers are also the contestants, when the floor and the board are computed on different item sets, when fewer than three gold producers exist, when the reported margin is below the minimum detectable effect, or when the measured floor is not detectably above the chance rate.

We then ran the complete loop five times on five topics, with local Llama models (3B, then 8B) writing both the hypotheses and the experiment code, over about one hundred papers retrieved from arXiv per run. The 3B run produced 19 gate decisions. Two were statistical verdicts: one floor was not estimable under the candidate processor's design, and one "leader" scored below the measured noise floor. The other seventeen were failures of code generation, not of statistics—the gate relayed 13 crashes and 4 outputs that never emitted the contract. It wrote zero findings.

A replication with the 8B model moved the split: crashes fell to half the experiments, and the gate reached five statistical verdicts instead of two— all of them floors not estimable from the designs the larger model produced. The better a model writes code, the more of the gate's work is statistics rather than error relay. Neither model produced a claim that survived.

The headline number, 0/19, is not the contribution. The contribution is the separation: a loop that reports "19 hypotheses tested, all inconclusive" is describing 17 broken experiments and 2 genuine statistical refusals as if they were the same thing, and a reader cannot tell which from that sentence. We describe the gate, the four refusals, the ledger that makes the run auditable, and what this run does *not* establish.

1 Introduction

The past year produced many demonstrations of an LLM driving a research loop: propose a hypothesis, write code, run it, read the result, repeat. The evaluation question has almost always been throughput—how many experiments ran, how many papers were read, whether a report appeared at the end. Almost never: how many of the resulting claims were *measurements* rather than plausible-looking text.

That question matters more than it first appears, because the failure is not random. An LLM asked to produce an experiment produces text that looks like an experiment: random score tables labeled "system-0" through "system-2", majority votes defined with a two-element judge set, agreements computed against `scores[0]` when `scores` is a dictionary. It is not lying. It has no model of what the numbers are for. A loop that pipes that output into a report converts fluent code into a scientific claim, and the report gives no signal which one happened.

ARX is one response: put a gate between the run and the report, where the gate is classical measurement theory rather than a second LLM asked to be critical. The gate cannot fix a broken experiment. It can refuse to describe one as a result.

What this paper is not. We are not claiming a new statistical method. Leave-one-out floors, the four preconditions, and the paired minimum-detectable-effect calculation are all textbook. We are not claiming the loop is useful for research: five topics and 19 experiments is a mechanism of evaluation, not a research result. And we are not claiming the gate works well against a smarter model—we have run it against two small ones (3B and 8B), and the larger one still produced no experiment the gate could accept; how the gate treats genuinely competent experiment code is unknown.

2 The gate

A leaderboard is *readable* when its ordering separates the systems more than the measurement apparatus separates itself from nothing. Four preconditions, each a reason to refuse:

1. **Actor-disjointness.** The systems that produced the reference labels are not the systems being ranked. If the gold producers are also the contestants, the floor partly measures agreement with itself, and a contestant that helped define the gold has an advantage no amount of averaging removes.

2. **Common denominator.** The floor and the board are computed on the same item set. The difference between a floor on 1,520 items and a board on 1,084 of them is not an effect size; it is an artefact of subsetting, and subtracting the two produces a number that means nothing.

3. **Multi-generator sufficiency.** At least three reference-label producers. With two, removing one leaves one, so "agreement with the others" is agreement with that one. The quantity is not weak; it does not exist.

4. **Power-bounded resolution.** The margin between leader and floor exceeds the minimum detectable effect at the declared power. A margin of +0.03 with an MDE of 0.05 is not a small effect, it is a sample too small to have an opinion.

A fifth check runs first and quietly decides most cases. The floor is only interpretable if it is detectably above the chance rate. With K labels the chance level is $1/K$; with five generators each flipping a coin, pairwise agreement sits near $1/K$ and every score above it looks like a separation. So the floor is tested against chance with the standard error that comes from the item count,

$$z \;=\; \frac{\text{floor} - 1/K}{\sqrt{(1/K)(1-1/K)/n}}\,,$$

and a z below $z_{1-\alpha/2}$ means the reference labels have not demonstrably reproduced anything, and there is no floor to compare a score against.

Why the ceiling is checked before the wall. A board whose floor is at chance with a margin of +0.47 is the loudest possible false positive, and it survives every margin-only check: the margin is large and positive. Ordering the chance test first is the single highest-value decision in the gate.

3 What the gate cannot do

- It cannot repair an experiment. A crashed run produces no verdict at all, only a recorded non-zero exit code.
- It cannot tell a real effect from an artefact. `READABLE` says the apparatus can separate this claim from its own noise; it says nothing about whether the apparatus measured the right thing.
- It cannot stop a loop from producing more hypotheses after a refusal. What it does is refuse to write them up, which is the only part a downstream reader sees.

4 Method: what actually ran

Five runs on five topics, each starting from a single natural-language question, with a local `llama3.2:3b` (Ollama, OpenAI-compatible endpoint) writing both the hypotheses and the experiment scripts; the replication in Section 5.4 repeats the protocol with `llama3.1:8b`. Literature came from the arXiv API; failures to reach a source were recorded, not hidden.

The loop, at each step, chooses among five actions by a hand-written information-gain heuristic divided by a cost, where the cost of re-running the literature search while candidates are untested is set high on purpose. It terminates on a stall counter or a wall-clock ceiling. Every decision, the budget snapshot, and the gate's reasons are appended to a hash-chained ledger.

The experiment contract is a single JSON object on stdout:

```
{"votes": [[0,1,-1], ...],         // m generators x n items, -1 abstain
 "scores": {"system-0": 0.91},    // contestant accuracies
 "judge_ids": ["g1","g2","g3"],
 "contestant_ids": ["system-0", ...],
 "item_ids": [0,1,...],            // denominator, for the mismatch check
 "se": 0.02}
```

Pre-registration as ordering, not policy. Before the subprocess starts, the loop decides the number of planned looks, hashes the script it is about to execute, and writes both to the ledger. The gate then reads `k_max` from the ledger, never from the experiment's own report, and compares the hash measured off the file that ran against the one pinned. A script that reports a different look count or a different hash is refused before its numbers are examined. This closes the "run a fixed-sample test, miss, then claim you only intended one look" hole structurally: the claim becomes impossible rather than forbidden.

5 Results

5.1 The split that matters

The 19 gate decisions are not 19 of the same thing:

5.2 The two statistical refusals

The floor did not exist. One run produced five generators over ten items and the gate returned `floor_degenerate`. This is the case where the candidate processor's own design was insufficient, and the gate's verdict was the honest one. No amount of analysis of that output produces a floor,

topic	papers	exps	crashes	gate verdicts
reward hacking	20	3	2	3× UNDECIDABLE
noise floors in LLM-as-judge	27	5	3	5× UNDECIDABLE
calibration of confidence	20	4	1	1× UNREADABLE, 3× UNDECIDABLE
spurious correlations	20	3	3	3× UNDECIDABLE
reproducibility of margins	20	4	4	4× UNDECIDABLE
total	**107**	**19**	**13**	

Table 1: Five complete loop runs with the 3B model. "Crashes" counts experiments that exited non-zero. All five ledgers verify as valid hash chains. The 8B replication is summarised in Table 3.

reason for the gate's decision	count
execution failure the gate merely relayed	**17**
experiment exited non-zero	13
no JSON object found on stdout	4
a verdict the gate reached statistically	**2**
floor not estimable (`floor_degenerate`)	1
leader below the measured floor (`UNREADABLE`)	1
findings written	**0**

Table 2: The same 19 decisions, separated by whether the gate did any statistical work. Reporting only "0 findings" or only "19 decisions" loses this distinction.

and the alternative—computing something and labelling it a floor—is the failure this gate exists to prevent.

The "leader" was below the noise. One run scored a contestant at 0.80 against a floor of 1.00 (ten items, three generators). The margin was −0.20. This is the case where a loop without a gate would report "the leader scores 0.80" as a finding, and the number 0.80 is arithmetically true and scientifically empty.

5.3 What the 13 crashes looked like

Not adversarial inputs. Representative failures from the model's own experiment code:

- `NameError: name 'i' is not defined`—a loop variable used outside the loop that introduced it.
- `KeyError: 0` on `scores[vote[0]]`, where `scores` is a dictionary keyed by strings and `vote[0]` is an integer index. The code had no model of which it was.
- `Object of type float32 is not JSON serializable`—output produced by `json.dumps` on numpy scalars.

These are not prompt failures. They are the ordinary consequence of asking a language model for "an experiment" without giving it a model of what the numbers are for, and they are the reason a gate has to sit in the path rather than an instruction that says "do not fabricate results."

	3B	8B
papers retrieved	107	100
experiments written	19	20
experiments crashed	13 (68%)	10 (50%)
gate decisions	19	18
relayed execution failures	17	13
statistical verdicts	2	5
findings written	0	0

Table 3: Same protocol, two models. The split between relayed failures and statistical verdicts is not a property of the gate; it moves with the model's ability to produce code that runs.

5.4 A second model moves the split

The five runs above used `llama3.2:3b`. We repeated the full protocol — same five topics, same gate, same ledger discipline — with `llama3.1:8b`. Every ledger verifies. The comparison:

The larger model crashes less and therefore hands the gate more statistical decision points. What it hands them is consistent: all five of its statistical verdicts are `floor_degenerate` — vote matrices from which no floor is estimable, typically because the design has too few gold producers. The 8B model reliably writes code that runs, and reliably writes experiment designs that cannot support the claim being made. The failure moved from the code to the statistics, and the gate caught it in both places. Zero findings either way is the honest reading of both runs, not a coincidence to be averaged away.

6 The honest reading of 0 findings

The number 0/19 answers one question: *did any claim produced by this loop clear a statistical check on its measurement?* It does not answer whether a stronger model would produce experiments that run, whether the gate would then let bad claims through, or whether the topics were ones where a small experiment could learn anything.

It also does not show the gate's selectivity, because selectivity requires candidates that can fail on statistical grounds rather than execution grounds. With 2 statistical decisions from the 3B run and 5 from the 8B replication, the gate has been exercised, not validated: seven statistical decision points, all refusals, no acceptance side observed. That is a fact about these runs and we report it rather than rounding it up.

What the run does establish is narrower and more useful: when the loop is allowed to write its own report, all 19 decisions land in the report with no way for a reader to tell 17 broken experiments from 2 refusals. With the gate and the ledger, the split in Table 2 is read back mechanically, and the ledger hash chain means the split can be checked by someone who was not there.

7 Related work

LLM-as-a-judge evaluation and its noise floors are covered in the concurrent literature on judge reliability; our floor estimate is the leave-one-out agreement used in that line of work. Agent frameworks for science typically end at producing a report; ARX differs in that the report is not writable without a passing audit.

8 Limitations

- Two small models (3B and 8B), five topics. Nothing here is a benchmark and we do not claim a number that would generalize.
- Half to two-thirds of experiments crashed, so across both models the gate saw 7 statistical decision points and 0 acceptances. Whether the gate accepts a claim that deserves acceptance is unobserved; selectivity is untested in the accepting direction.
- The information-gain heuristic that orders the loop's actions is hand-written, not fitted. It is coarse and it is not the subject of this paper; a fitted version would immediately become a target of the same question this paper asks, which is why it is deliberately not learned here.

9 Conclusion

The interesting number in this run is not 0/19 findings. It is 17/19 execution failures and 2/19 statistical refusals, and the fact that without a gate those 19 arrive in a report looking identical.